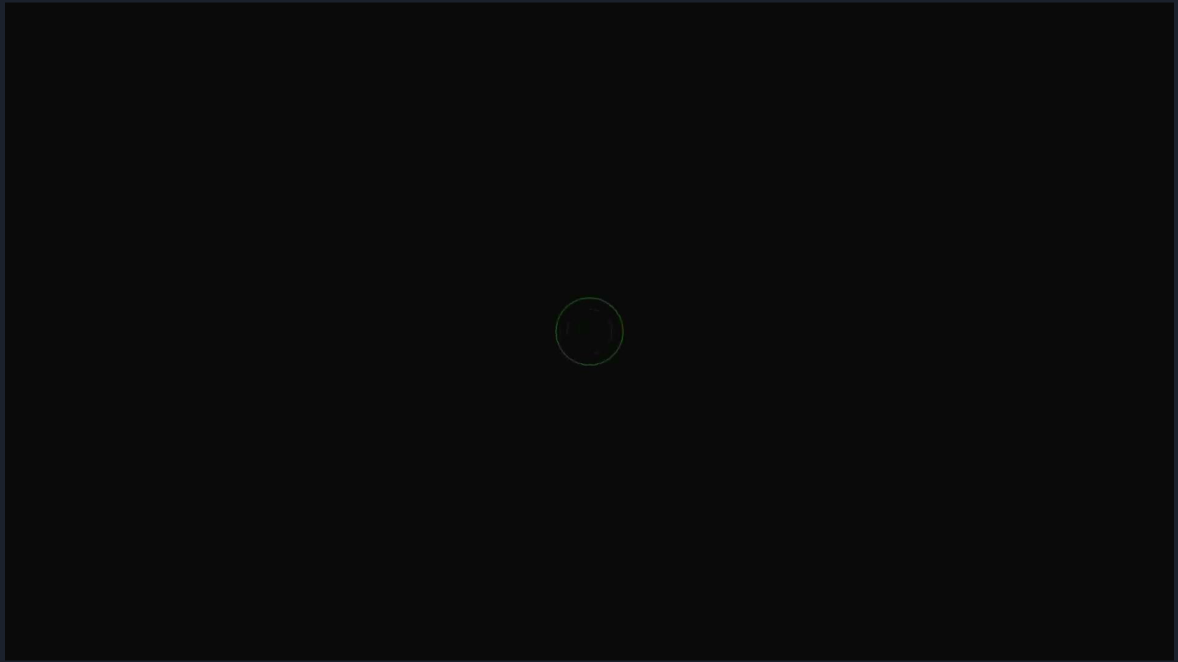
playerdata™ x IU INDY

Final Presentation

4/29/2026

TESM-T582: Applied Sport Event Research

Happy Birthday, Buddy!!



Overview

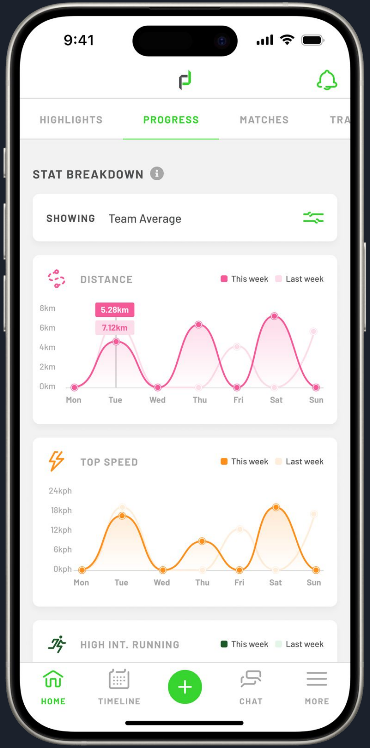
This project delivers a scalable, comparative benchmarking engine for PlayerData, enabling coaches and organizations to interpret athlete performance in context.

- Built a cross-functional pipeline integrating engineering, science, and analytics.
- Produced synthetic data sets, statistical norms, and multi-platform visualizations.
- Provides a blueprint for in-app enhancement.

The Current Context Gap

PlayerData users can see what an athlete did, but not how it compares.

- ➢ Coaches lack percentile benchmarks.
- ➢ Organizations lack role-specific or age-specific norms.
- ➢ The project goal: bridge the context gap with Comparative Intelligence.



Project Scope & Governance

A governance prototype ensured accuracy and velocity.

- Centralized Google Drive “Control Room.”
- GitHub Kanban for sprint orchestration.
- Strategic check-ins on March 4 and April 8 guided technical builds.

Data Architecture & Engineering

Dataset: 7,868 sessions of 281 athletes.

- ➢ Applied a 70-minute active_minutes floor to remove low-intensity sessions.
- ➢ Identified Session Load as the primary composite intensity metric.
- ➢ Built two synthetic datasets (10,000 sessions each) to validate population shifts.

Engineering Innovations

- ➢ Used the Men’s DII and Women’s DIII synthetic populations to develop pipeline for metric creation prior to receiving finalized data.
- ➢ Prototyped a Conversational AI Agent that returns contextualized natural-language insights.

Data Science Focus

PlayerData feedback narrowed the scope to four high-impact metrics:

- Total Distance
- Max Speed
- Session Load
- Sprint Events

Methodology:

- Weighted by active minutes.
- Filtered for elite training (>70 min).
- Aggregated by age (18–21) to produce normative means & standard deviations.
- Data Imbalance: Age group size inequities reduce generalizability.

Benchmarks: Total Distance (meters)

Athlete Relative Age	Session Count	Average	Standard Deviation	90th Percentile	10th Percentile	Max.	Min.
18	1472	3853.86	2422.99	7018.37	454.97	11982.59	113.00
19	306	5296.34	1871.28	7706.43	2958.24	9200.14	134.60
20	1086	4063.22	1943.84	6761.51	1580.41	10785.97	100.02
21	334	3641.52	1603.80	5390.36	1683.18	9777.45	106.04

- Standard deviation for 18 year olds is higher than the rest, despite their increased sample size. Potential variation in background and adjustment to collegiate athletics could contribute to varying comfort and performance levels.

Benchmarks: Session Load

Athlete Relative Age	Session Count	Average	Standard Deviation	90th Percentile	10th Percentile	Max.	Min.
18	1472	544.76	412.95	1093.27	37.17	2313.27	4.30
19	306	701.18	301.25	1010.61	369.86	2685.77	7.61
20	1086	542.82	315.60	955.34	189.57	2772.07	4.93
21	334	543.40	318.65	988.64	210.51	1986.89	6.52

- Sophomore peak: 19 year olds had noticeably higher session loads on average compared to the other age groups.

Benchmarks: Max Speed (kilometers per hour)

Athlete Relative Age	Session Count	Average	Standard Deviation	90th Percentile	10th Percentile	Max.	Min.
18	1472	20.30	5.86	26.28	9.09	31.65	4.18
19	306	21.89	3.91	25.96	17.83	28.88	5.27
20	1086	21.55	3.95	25.51	17.34	30.73	4.54
21	334	21.76	4.13	26.63	16.21	33.89	5.78

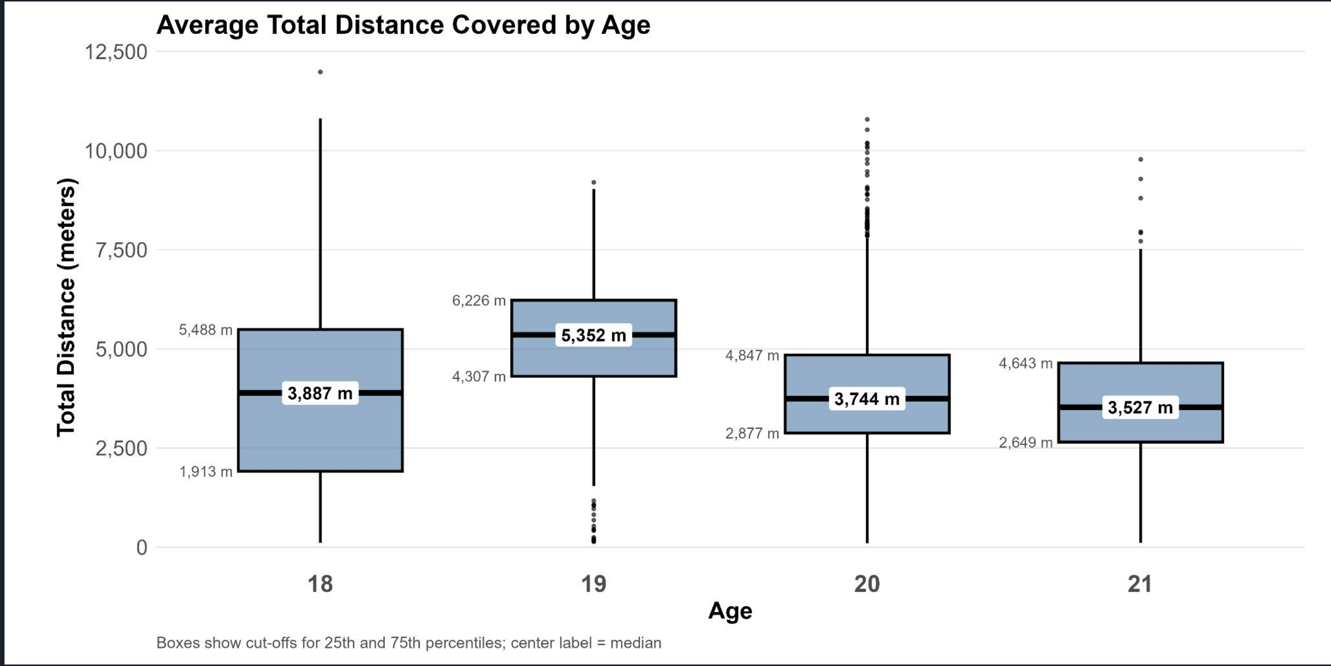
- Stability: Both average and 90th percentile max speed remained fairly stable across age groups. 18 year olds had a slightly lower average.

Benchmarks: Sprint Events

Athlete Relative Age	Session Count	Average	Standard Deviation	90th Percentile	10th Percentile	Max.	Min.
18	1472	1.28	2.29	4	0	16	0
19	306	1.67	2.44	5	0	15	0
20	1086	1.29	2.39	4	0	20	0
21	334	2.12	3.70	7	0	26	0

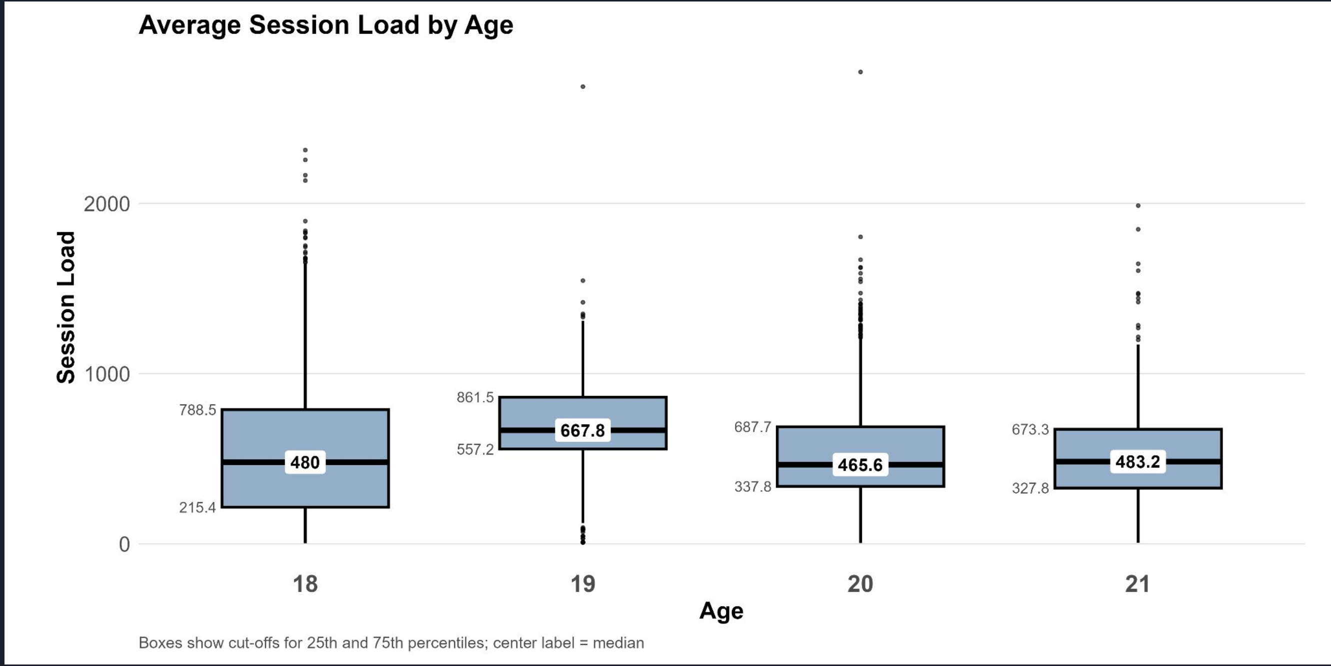
- Settling into the game: 21 year olds had a higher average and showed more volatility with a standard deviation higher than the other age groups.

Visualization Insights: Total Distance



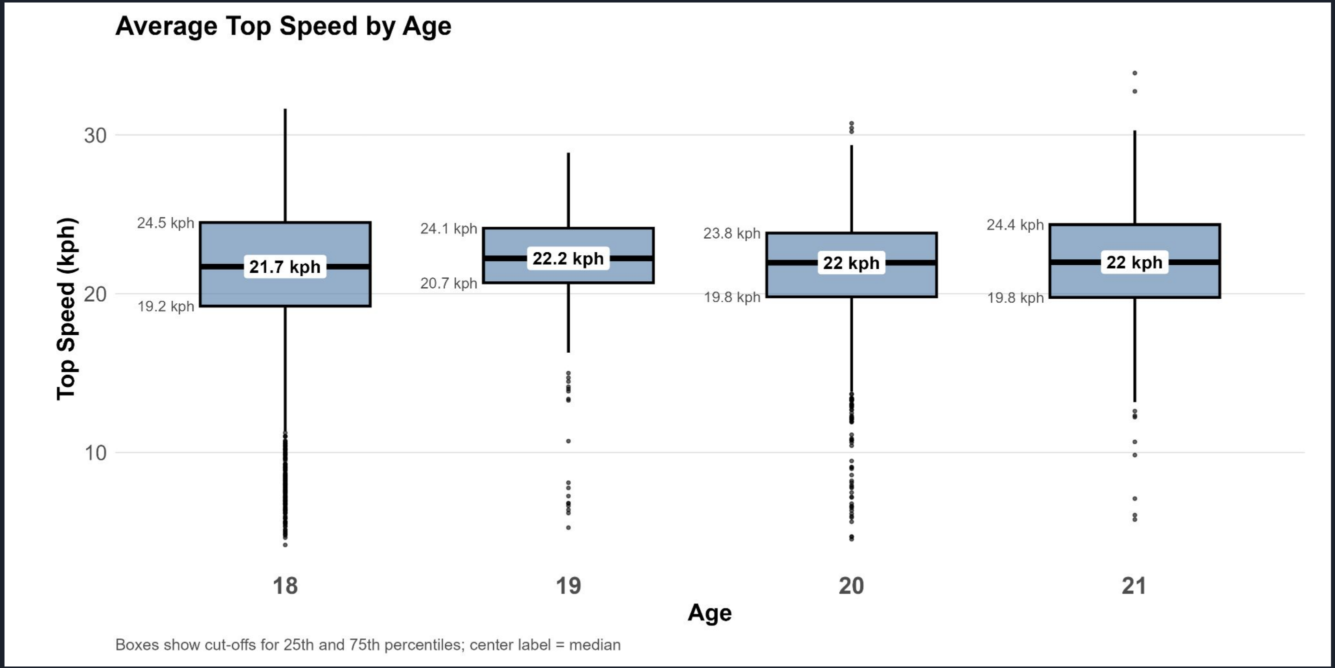
- **18 Year Old Median:** 3,887 m
- **19 Year Old Median:** 5,352 m
- **20 Year Old Median:** 3,744 m
- **21 Year Old Median:** 3,527 m

Visualization Insights: Session Load



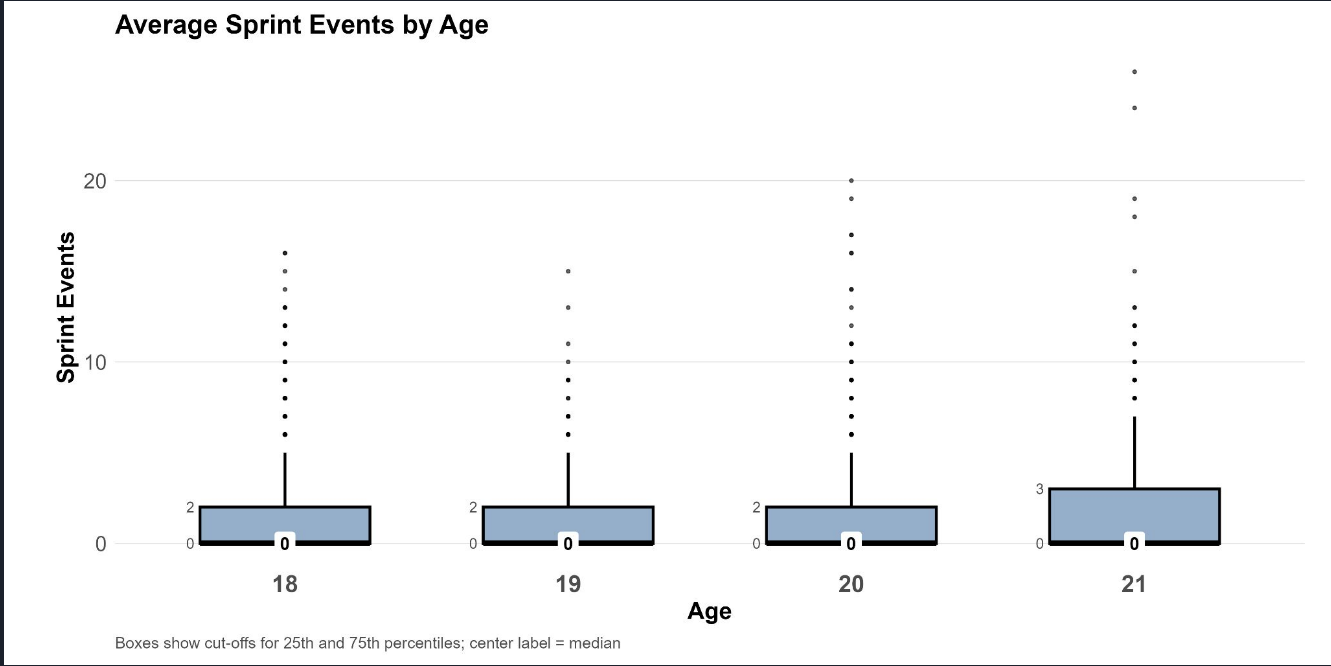
- ➢ **18 Year Old Median**: 480.0
- ➢ **19 Year Old Median:** 667.8
- ➢ **20 Year Old Median**: 465.6
- ➢ **21 Year Old Median:** 483.2

Visualization Insights: Top Speed



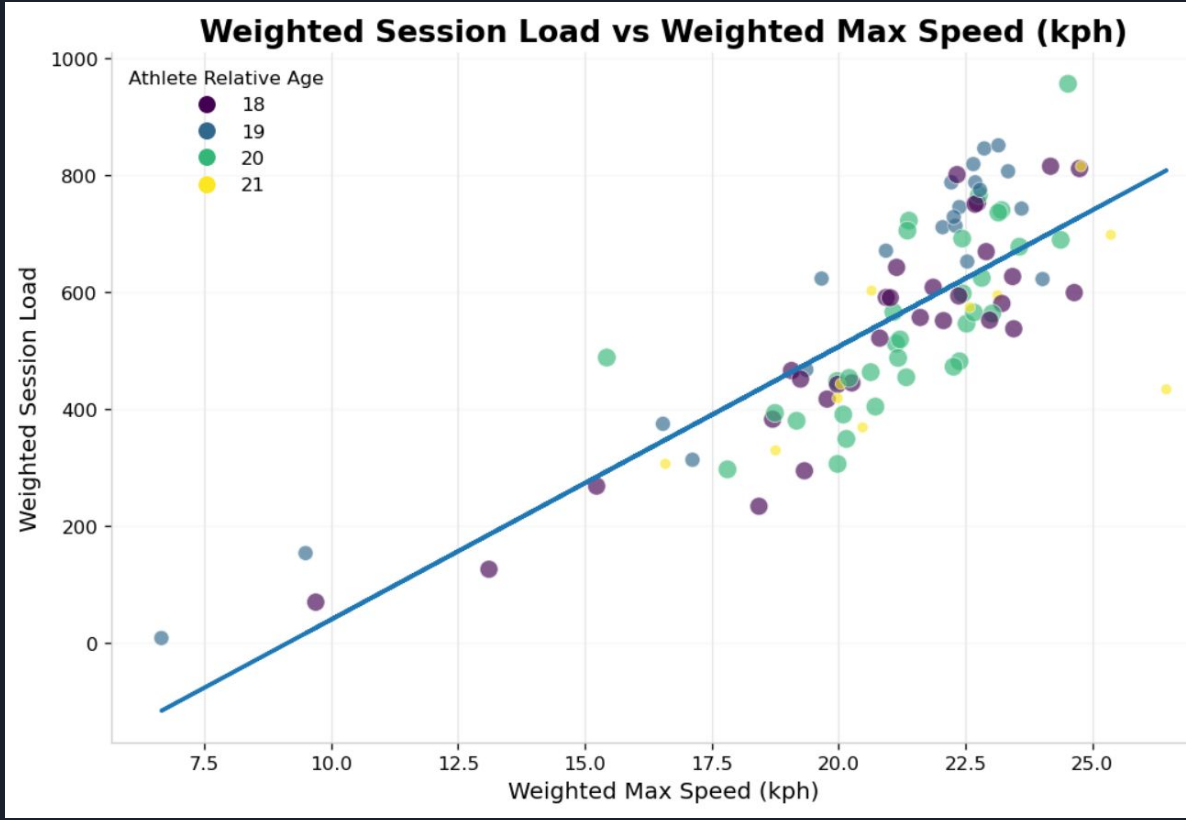
- ➢ **18 Year Old Median**: 21.7 kph
- ➢ **19 Year Old Median**: 22.2 kph
- ➢ **20 Year Old Median**: 22.0 kph
- ➢ **21 Year Old Median**: 22.0 kph

Visualization Insights: Sprint Events



- ➢ **18 Year Old Median:** 0 Sprint Events
- ➢ **19 Year Olds Median:** 0 Sprint Events
- ➢ **20 Year Old Median:** 0 Sprint Events
- ➢ **21 Year Old Median:** 0 Sprint Events

Feature Correlation



- ➢ Nearly linear positive correlation.
- ➢ Validates Session Load as a proxy for high-intensity sprinting.

Product Application for Coaches

How coaches use the benchmarks:

- ➢ Identify 90th-percentile athletes → reduce load to prevent injury.
- ➢ Identify 10th-percentile athletes → increase conditioning.
- ➢ Integrate with AI agent for plain-text, on-demand insights.

Limitations

- ➢ Limited temporal depth → cannot confirm whether the "Sophomore Peak" is biological or data--specific.
- ➢ Limited multisport data → cannot generalize beyond women's soccer.

Future Research

- ➢ Add positional and session-type context.
- ➢ Build season-over-season tracking for player evolution.
- ➢ Expand to additional sports.
- ➢ Implement production-grade data lineage.
- ➢ Creation of a "Rolling Load" Metric for teams.

Future Research & Roadmap

- Add positional and session-type context.
- Build season-over-season tracking for cohort evolution.
- Expand to additional sports.
- Implement production-grade data lineage.
- Creation of a Rolling Load Metric for teams

Product Demo